// generative instrument

Donut_OS

diceFriend / quick start

Two voices walk a wrapping pitch grid through six tunings, folding, granular memory and feedback.

START: USB-C power + 3.5 mm stereo audio out. Begin with the listening level low.

T1 taps tempo. T2 pressure adds brightness and grain intensity. T3 pressure lengthens decay.

// SIX KNOBS / TWO LAYERS

P	NORMAL	SHIFT
1	Scale: 6 tuning maps	Density: audible-step probability
2	Interval span	Chaos: grain / register instability
3	Width: 2-32 cells	Mutation: direction / spectrum
4	Height: 2-32 cells	Feedback: motion / fold / colour
5	Bias: X / Y motion	Inertia: retain axis / direction
6	Root: C1-C4	Route: body / texture / decay

// TUNINGS / ROUTE

Tunings: 12 EDO / 19 EDO / 31 EDO / Hexany A / Hexany B / 5-limit JI. ROUTE selects how grid position, internal chaos and CV2 reach body / folding, texture / FM or decay.
SHIFT: up = second layer; down = normal. Pickup: after changing layers, using DICE or receiving CC, move a knob to take control.

// DICE / LOOP

Tap DICE: new world and immediate strike. SHIFT + tap DICE: related mutation preserving scale, root, grid and route. Hold DICE 450 ms: loop up to 8 recent events; release to continue from the recalled position. SHIFT + DICE 10 s: bootloader.

// CLOCK / TRANSPORT

CV > MIDI > internal. MIDI may take over 20 s after CV; internal returns after 30 s without external activity. MIDI Clock is 24 PPQN. Start / Continue run; Stop silences MIDI-clocked motion. Note On can still strike after Stop.

// CV IN

CV1: priority step / clock. CV2: continuous ROUTE modulation or secondary trigger / clock; CV1 blocks extra CV2 steps. Banana: 0-9 V; common GND.

// USB MIDI

IN ch 1: Note On forces a step; note number does not retune. CC20-31 = panel, CC74 / Aftertouch = expression. OUT: voices A / B on ch 1 / 2 with Pitch Bend, Note On / Off and internal / CV Clock.

// FIRST PATCH

Choose 19 EDO, medium WIDTH / HEIGHT, high DENSITY and low CHAOS / MUTATION / FEEDBACK. Tap T1 twice. Raise CHAOS, then hold DICE when an eight-step phrase appears.

// LEDS

Layer LEDs show NORMAL / SHIFT. Lower LEDs: step activity; CV2 or loop blink; left audio; right audio.